Python has built in data types

list
tuple
set
dict
String

} class

Using OOPs we can create our own data type in python

Magic Method - (Dunder method)

① Magic methods are special methods which triggred automatically at specific situation.

```
def fun_name():
    ____
    ____
```

↑ fundefinition ← method

fun.name() ← fun call

```
class Student:
    def __init__(self):
        self.name = "rahul"
        self.age = 25

Student1 = Student()
```

① __init__ :- when we create an object

② __str__ :- when we print an object

```
print(Student1)
```

③ __add__ :- when we add two objects

```
obj1 + obj2
```

④ __sub__ :- when we subtract two objects

```
obj1 - obj2
```

⑤ __mul__ :- when we multiply two objects

obj1 * obj2

⑥ __truediv__ :- when we divide two objects

obj1 / obj2

Built in data types —

list, tuple, set, dict, string

eg. class list :

def append(self):

def extend(self):

list1 = list() ← obj (instance)

list1.append(10)

print (list1) ⟶ [10]

Fraction — $\frac{1}{2}$, $\frac{4}{9}$

Fraction Rules :-

$$\frac{A}{B} \times \frac{C}{D} = \frac{AD + BC}{BD}$$

$$\frac{A}{B} \times \frac{C}{D} = \frac{AD - BC}{BD}$$

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

$$\frac{A}{B} \div \left(\frac{C}{D}\right) = \frac{A}{B} \times \frac{D}{C} = \frac{AD}{BC}$$

eg.

$$\frac{1}{2} \times \frac{3}{4} = \frac{1 \times 4 + 2 \times 3}{2 \times 4}$$

$= \frac{4+6}{8} = \frac{10}{8}$

$\frac{1}{2} \times \frac{3}{4} = \frac{1\times4 - 3\times2}{2\times4} = \frac{4-6}{8}$

$= \frac{-2}{8}$

what is fraction :-

numerator

denominator

eg. $\frac{1}{2}$ → fraction

0.5 → decimal ✓

fraction does example

$\frac{1}{2}$, $\frac{7}{11}$

fraction1 fraction2

self other

$\frac{1}{2} + \frac{7}{11} = \frac{1\times11 + 2\times7}{2\times11} = \frac{11+14}{22}$

new_num = (self.num * other.den) + (self.den * other.num)

new_den = self.den * other.den

$= \frac{25}{22}$ ✓

$\frac{1}{2} - \frac{7}{11} = \frac{1\times11 - 2\times7}{2\times11} = \frac{11-14}{22}$

$= \frac{-3}{22}$

$\frac{1}{2} \times \frac{7}{11} = \frac{1\times7}{2\times11} = \frac{7}{22}$

$\frac{1}{2} \div \frac{7}{11} = \frac{1}{2} \times \frac{11}{7} = \frac{1\times11}{2\times7} = \frac{11}{14}$

num 1 den 2 + num 7 den 11 $= \frac{1\times11 + 2\times7}{2\times11}$

```
def __add__(self, other):
    new_num = (self.num * other.deno +
               self.den * other.num)
    new_den = (self.den * other.den)
    return fraction(new_num, new_den)
```

return Fraction(

print(fraction1 + fraction2)